

Objectives

Define Actuators

Differentiate between motion Actuators & non motion Actuators

Produce different tones using piezo buzzer

Control Active Buzzer

Actuators

Activate Windows
Go to Settings to activate Windows.

Actuator

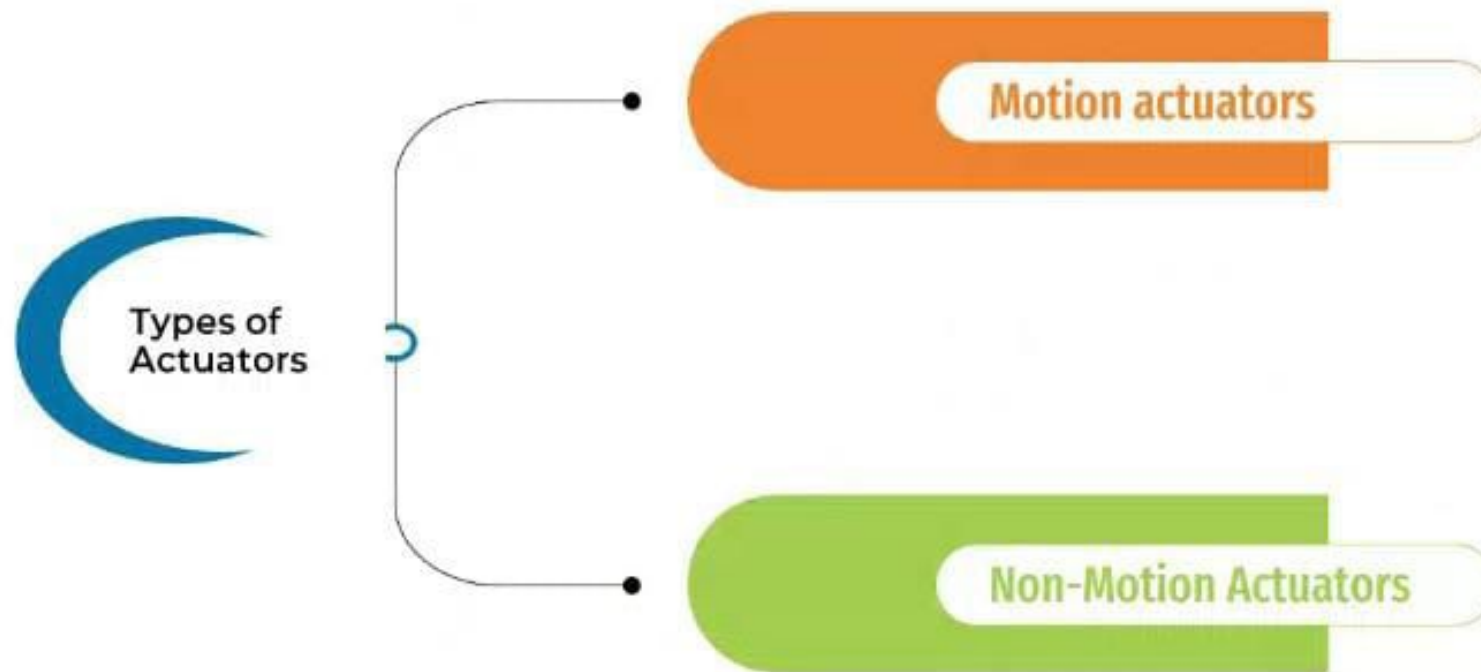
An actuator is a device that moves or controls a certain mechanism.

An actuator turns a control signal into a mechanical action such as an electric motor.

Actuators may be based on pneumatic, electric, thermal or mechanical means, but are increasingly being driven by software.



Types of Actuators



Non Motion Actuators

We can list the LCD, LED, RGB, buzzer, laser transmitter, and IR transmitter. All of these electric element execute electrical actions, like emitting light, or signal.



Active Buzzer

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Active Buzzer

A **buzzer** is a small speaker that uses the piezo element to create a sound. the piezo element is a very sensitive material that reacts with electric waves. This will create sound waves due to the vibration of the piezoelectric element.

A piezoelectric speaker requires a `tone()` function to work:

- 1- `tone( pin number, frequency, Time in ms);`
- 2- The pin number and frequency are essential for the piezo to work.
- 3- The "time in the second" is optional



Tone () function

In this cases, we will consider having the buzzer connector to pin 10 and we have defined the pin.

Case	programming	Output
1	<pre>void loop(){ tone (10 , 3000); }</pre>	the buzzer will produce a tone on 3000 Hz as long as the Arduino is ON
2	<pre>void loop(){ tone (10 , 3000); delay (1000); }</pre>	The buzzer will produce a tone on 3000 Hz as long as the Arduino is ON because there is only this tone.
3	<pre>void loop(){ tone (10 , 3000, 500); }</pre>	The buzzer will produce a tone on 3000 Hz as long as the Arduino is ON because there is only this tone.

Tone () function

4	<pre>void loop(){ tone (10 , 3000, 500); delay(1000); }</pre>	The buzzer will produce sound for 500 ms then it will stop for only 500 ms. this is because the program has only one time, so both comments will start at the same time.
5	<pre>void loop(){ tone (10 , 3000, 500); tone (10 , 4000, 500); }</pre>	The Arduino will produce tone on 3000 Hz then 4000 tones, which will create a weird sound
6	<pre>void loop(){ tone (10 , 3000, 500); delay(1500); tone (10 , 4000, 500); delay(1500); }</pre>	The Arduino will produce tone on 3000 Hz for 0.5 S then delay 1 S Then 4000 tone for 0.5 S. then delay 1 S

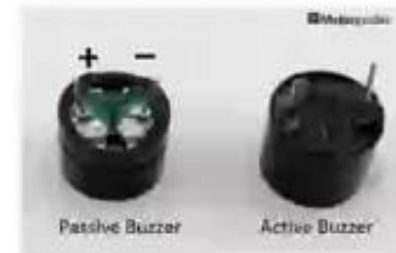
Tone () function

The limits of tone () function:

- 1- you can't use PWM on pin 3 and 11 while using tone()
- 2- The minimum hertz is 31
- 3- You can only use tone () function on one pin.



Active Vs Passive Buzzer



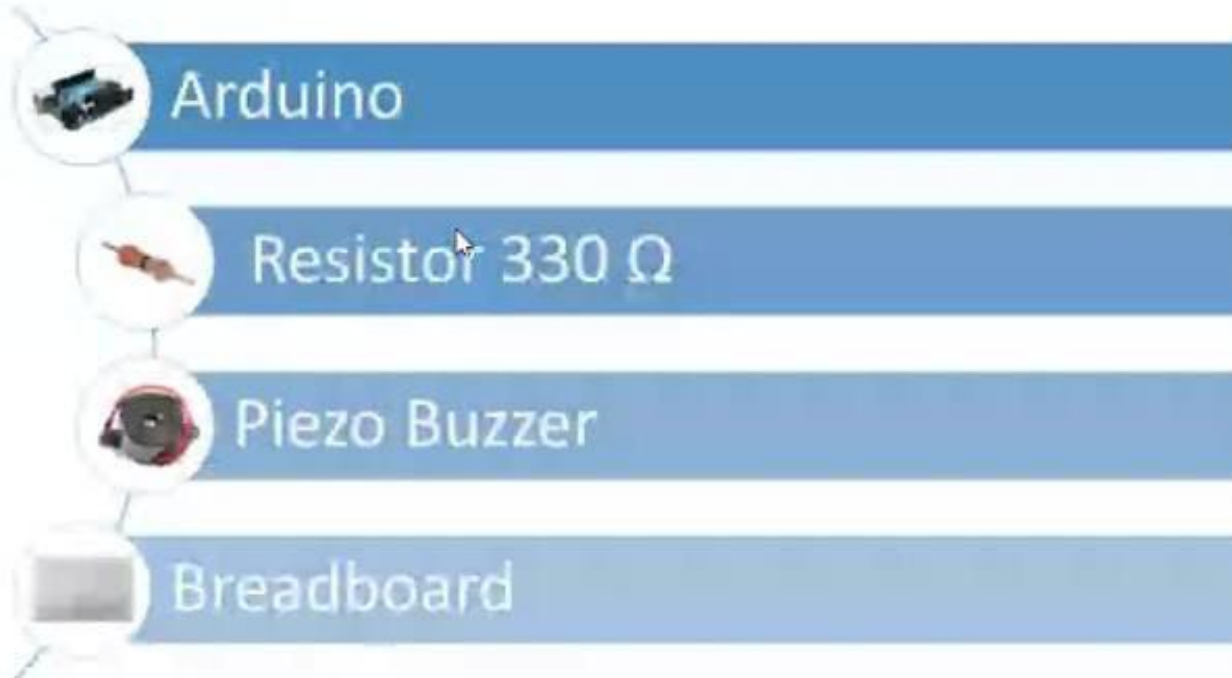
The core difference between active and passive buzzers is that active buzzers have a built-in oscillator to produce a single fixed tone from a DC voltage, while passive buzzers require an external AC or square-wave signal to generate different frequencies and musical notes.

- **Active Buzzer:** Contains an internal driver circuit (oscillator). When you apply a steady DC voltage (like 5V), it automatically vibrates and makes a constant sound.
- **Passive Buzzer:** Has no internal oscillator. It works like a tiny speaker and only clicks once when given steady DC power; it needs changing signals (like PWM waves) to create tones.

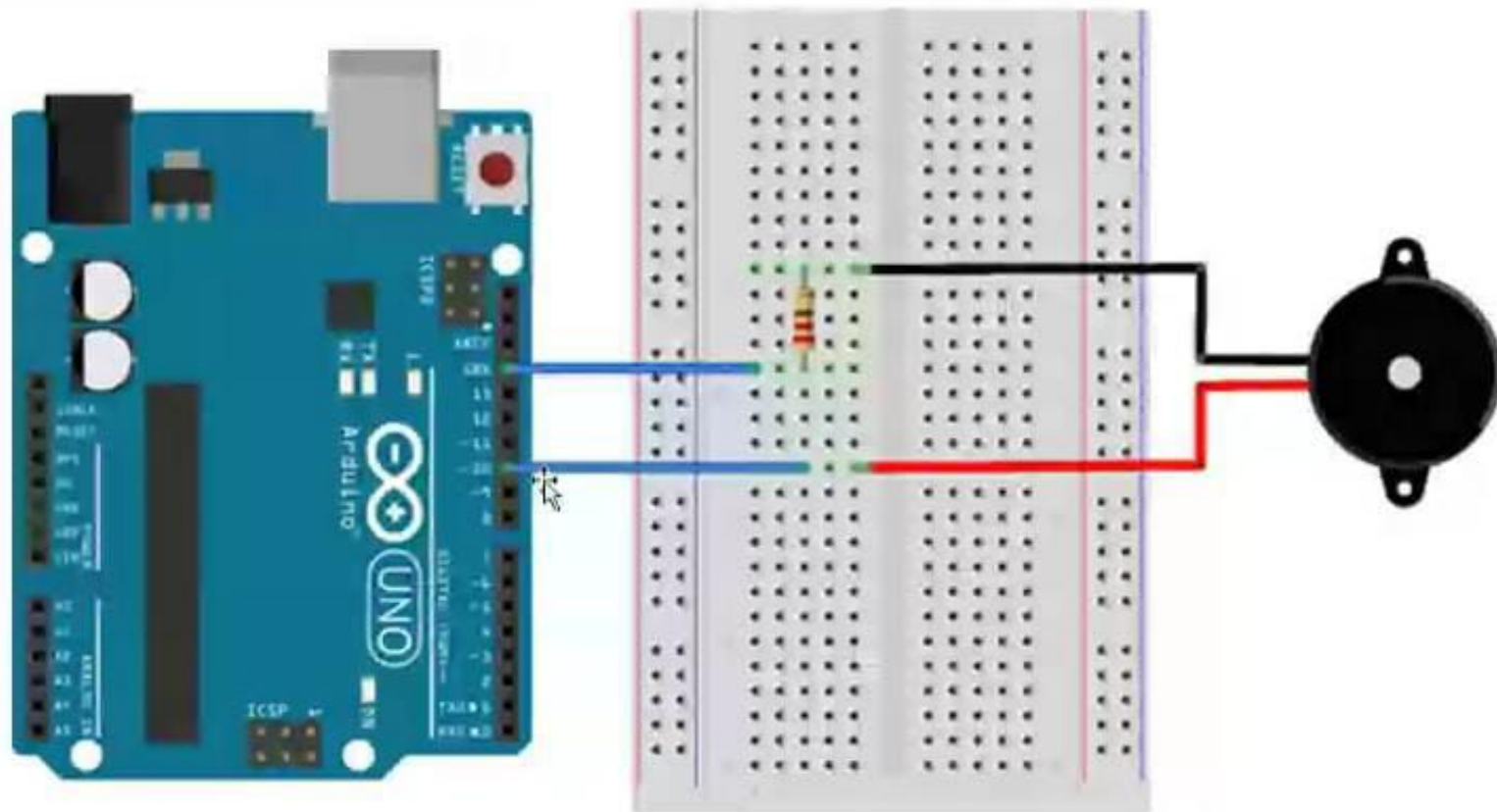
Passive buzzers provide full control over frequency and timing. They are suitable for applications where different tones or melodies are required. Active buzzers are simpler to use. They are suitable for alarms or status indicators where only a single fixed tone is needed.

Buzzer

Required Elements:

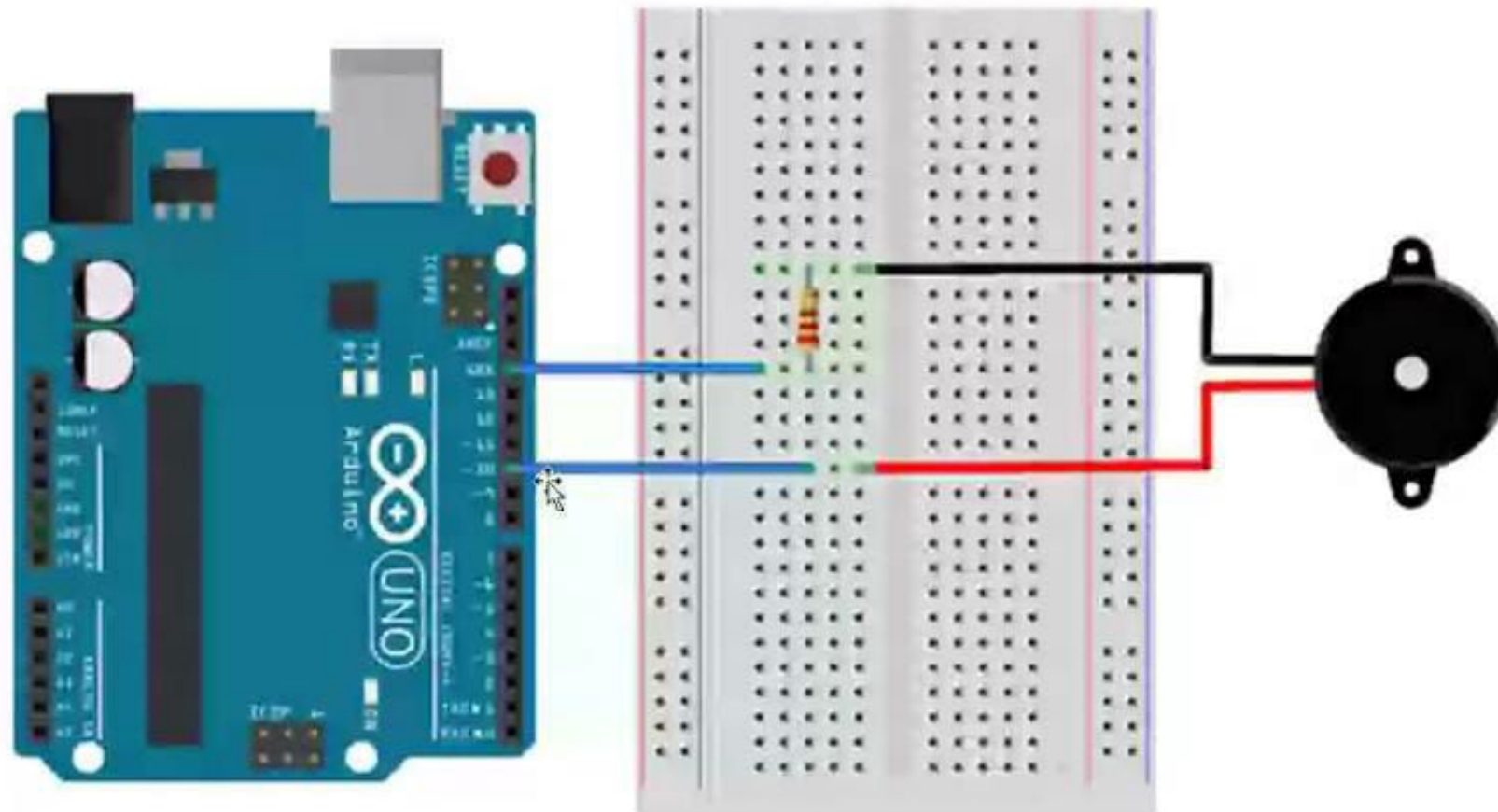


Board connection



fritzing

connection



fritzing

Activate Windows
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Programming

```
1 void setup() {  
2   // put your setup code here, to run once:  
3  
4 }  
5  
6 void loop() {  
7   // put your main code here, to run repeatedly:  
8   tone( 10, 1500, 500);  
9   delay(1000);  
10  tone( 10, 1000, 500);  
11  delay(1000);  
12 }
```




Arduino UNO

Upload

sketch_jul29b.ino

```
1 void setup() {  
2     // put your setup code here, to run once:  
3  
4 }  
5  
6 void loop() {  
7     // put your main code here, to run repeatedly:  
8     tone(10,500,500);  
9     delay(1000);  
10    tone(10,1000,500);  
11    delay(1000);  
12    tone(10,1500,500);  
13    delay(1000);  
14    tone(10,2000,500);  
15    delay(1000);  
16    tone(10,3000,500);  
17    delay(1000);  
18 }  
19
```

Programming steps

Part Three: Music scale

Tone	Frequency in Hz
Do4	261
Re 4	293
Me4	329
Fa4	349
Sol4	392
La4	440
Ce4	493
Do5	523

12-Introduction to Actuators - PowerPoint (.pptx) Picture...

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Programming steps

Part Three: Music scale

Tone	Frequency in Hz
Do4	261
Re 4	293
Me4	329
Fa4	349
Sol4	392
La4	440
Co4	493
Do5	523

dot

Slide 20 of 21 Arabic (Lebanon) Notes Comments 60%

sketch_jul29b | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino UNO

sketch_jul29b.ino

```
1 void setup() {
2   // put your setup code here, to run once:
3
4 }
5
6 void loop() {
7
8   // put your main code here, to run repeatedly:
9   //tone(pinnumber,frequency of sound, duration c
10  tone(10,500,500); //speaker connected to pin 10 w
11  delay(1000);
12  tone(10,1000,500); //speaker connected to pin 10
13  delay(1000);
14  tone(10,1500,500);
15  delay(1000);
16  tone(10,2000,500);
17  delay(1000);
18  tone(10,3000,500);
19  delay(1000);
20 }
```

Activate Windows
Go to Settings to activate Windows.

Output

Ln 17, Col 13: Arduino UNO no COM5 (not connected)

4 PM 7/2026

```

1 void setup() {
2   // put your code here
3
4 }
5
6 void loop() {
7   // put your code here
8   do4(500);
9   re4(500);
10  me4(500);
11  fa4(500);
12  sol4(500);
13  la4(500);
14  ce4(500);
15  do5(500);
16  delay(2000);
17
18  do5(500);
19  ce4(500);
20  la4(500);
21  sol4(500);

```

```
int BuzzerPin = 10; // Connect Buzzer to Arduino pin 10
```

```

int Si2=1975;
int La52=1864;
int la2= 1760;
int Sol52=1661;
int sol2=1567;
int Fa52=1479;
int fa2= 1396;
int Mi2= 1318;
int Re52=1244;
int re2= 1174;
int Do52=1108;
int do2= 1046;

```

```

// Low Octave
int si = 987;
int la5= 932;
int la = 880;
int Sol5=830;
int sol= 783;
int Fa5= 739;
int fa= 698;
int Mi= 659;
int Re5= 622;
int re = 587;
int Do5 =554;
int do = 523;

```

```

// define the notes
int rounda=0;
int roundp=0;
int white= 0;
int whitep=0;
int black=0;
int blackp=0;
int quaver=0;
int quaverp =0;
int semiquaver=0;
int semiquaverp=0;

```


Arduino UNO

int BuzzerPin = 10; // Connect Buzzer to Arduino pin 10

sketch_jul29c.ino

```
1 void setup() {  
2   // put your  
3  
4 }  
5  
6 void loop() {  
7   // put your  
8   do4(500);  
9   re4(500);  
10  me4(500);  
11  fa4(500);  
12  sol4(500);  
13  la4(500);  
14  ce4(500);  
15  do5(500);  
16  delay(2000);  
17  
18  do5(500);  
19  ce4(500);  
20  la4(500);  
21  sol4(500);
```

```
int Si2 =1975;  
int La2=1864;  
int La2= 1760;  
int Sol2=1661;  
int Sol2=1567;  
int Fa2=1479;  
int Fa2= 1396;  
int Mi2= 1318;  
int Re2=1244;  
int Re2= 1174;  
int Do2=1108;  
int Do2= 1046;  
  
// Low Octave  
int Si = 987;  
int LaS= 932;  
int La = 880;  
int SolS=830;  
int Sol= 783;  
int FaS= 739;  
int Fa= 698;  
int Mi= 659;  
int ReS= 622;  
int Re = 587;  
int DoS =554;  
int Do = 523;  
  
// define the notes  
int rounda=0;  
int roundp=0;  
int white= 0;  
int whitep=0;  
int black=0;  
int blackp=0;  
int quaver=0;  
int quaverp =0;  
int semiquaver=0;  
int semiquaverp=0;
```

sketch_jul29c.ino

```
1 void setup() {  
2   // put your  
3  
4 }  
5  
6 void loop() {  
7   // put your  
8   do4(500);  
9   re4(500);  
10  me4(500);  
11  fa4(500);  
12  sol4(500);  
13  la4(500);  
14  ce4(500);  
15  do5(500);  
16  delay(2000);  
17  
18  do5(500);  
19  ce4(500);  
20  la4(500);  
21  sol4(500);
```

```
int ReS= 622;  
int Re = 587;  
int DoS =554;  
int Do = 523;  
  
// define the notes  
int rounda=0;  
int roundp=0;  
int white= 0;  
int whitep=0;  
int black=0;  
int blackp=0;  
int quaver=0;  
int quaverp =0;  
int semiquaver=0;  
int semiquaverp=0;  
int bpm= 120;  
  
void setup(){  
  pinMode(BuzzerPin,OUTPUT);  
  
  black= 35000/bpm;  
  blackp=black*1.5;  
  white= black*2;  
  whitep=white*1.5;  
  rounda= black*4;  
  roundp= rounda*1.5;  
  quaver= black/2;  
  quaverp=quaver*1.5;  
  semiquaver= black/4;  
  semiquaverp=semiquaver*1.5;  
}  
  
void loop(){  
  tone(BuzzerPin,Mi,black);  
  delay(black+50);  
  tone(BuzzerPin,La,black);  
  delay(black+50);  
  tone(BuzzerPin,Si,black);  
  delay(black+50);  
  tone(BuzzerPin,Do2,black);
```

Output




```

1 void setup(){
2   // put your code here
3
4 }
5
6 void loop(){
7   // put your code here
8   do4(500);
9   re4(500);
10  me4(500);
11  fa4(500);
12  sol4(500);
13  la4(500);
14  ce4(500);
15  do5(500);
16  delay(2000);
17
18  do5(500);
19  ce4(500);
20  la4(500);
21  sol4(500);

```

```

void setup(){
  pinMode(BuzzerPin,OUTPUT);

  black= 35000/bpm;
  blackp=black*1.5;
  white= black*2;
  whitep=white*1.5;
  rounda= black*4;
  roundp= rounda*1.5;
  quaver= black/2;
  quaverp=quaver*1.5;
  semiquaver= black/4;
  semiquaverp=semiquaver*1.5;
}

void loop(){
  tone(BuzzerPin,Mi,black);
  delay(black+50);
  tone(BuzzerPin,La,black);
  delay(black+50);
  tone(BuzzerPin,Si,black);
  delay(black+50);
  tone(BuzzerPin,Do2,black);
  delay(black+50);
  tone(BuzzerPin,La,black);

  delay(2*white+50);

  tone(BuzzerPin,Mi,black);
  delay(black+50);
  tone(BuzzerPin,La,black);
  delay(black+50);
  tone(BuzzerPin,Si,black);
  delay(black+50);
  tone(BuzzerPin,Do2,black);
  delay(black+50);
  tone(BuzzerPin,La,black);

  delay(2*white+50);

  tone(BuzzerPin,Mi,black);

```

```
1 void setup() {  
2   // put your  
3  
4 }  
5  
6 void loop() {  
7   // put your  
8   do4(500);  
9   re4(500);  
10  me4(500);  
11  fa4(500);  
12  sol4(500);  
13  la4(500);  
14  ce4(500);  
15  do5(500);  
16  delay(2000);  
17  
18  do5(500);  
19  ce4(500);  
20  la4(500);  
21  sol4(500);
```

```
int whitep=0;  
int black=0;  
int blackp=0;  
int quaver=0;  
int quaverp =0;  
int semiquaver=0;  
int semiquaverp=0;  
int bpm= 120;  
  
void setup(){  
  pinMode(BuzzerPin,OUTPUT);  
  
  black= 35000/bpm;  
  blackp=black*1.5;  
  white= black*2;  
  whitep=white*1.5;  
  rounda= black*4;  
  roundp= rounda*1.5;  
  quaver= black/2;  
  quaverp=quaver*1.5;  
  semiquaver= black/4;  
  semiquaverp=semiquaver*1.5;  
}  
void loop(){  
  tone(BuzzerPin,Mi,black);  
  delay(black+50);  
  tone(BuzzerPin,La,black);  
  delay(black+50);  
  tone(BuzzerPin,Si,black);  
  delay(black+50);  
  tone(BuzzerPin,Do2,black);  
  delay(black+50);  
  tone(BuzzerPin,La,black);  
  
  delay(2*white+50);  
  
  tone(BuzzerPin,Mi,black);  
  delay(black+50);  
  tone(BuzzerPin,La,black);  
  delay(black+50);
```



Servo Motor

dot
LEBANON

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Objectives

Define the Servo Motor

Manage the Arduino IDE Library

Use Servo Motor in Arduino Projects

Rotate servo motor with push button

Control servo using potentiometer

Servo Motor

Servo Motor is a DC motor with a gear transmission and controlling chip. The servo motor main characters that should be checked.

- **Servo torque:** The servo torque decides how much the servo motor can hold.
- **Servo limiting angle:** servo limiting angle depends on the servo type. Some servo have 180° or 270° degrees as the limit angle.
- **Servo speed:** each servo has a certain rotating speed. This speed depends on the voltage we are supplying the servo with.



Servo motor function

The servo motor is a motor with a gearbox that increases the torque output. The servo motor can drive a certain angle (180° or 270° or 360°) in both ways. The servo motor will position depending on the signal it is receiving. Then the servo will keep this position as long as it is still receiving a signal.

Most of the servos have three pins:

1. The VCC pin, which is used to receive electricity
2. The GND pin, which is used to return the electricity
3. The control pin, which is used to decide the servo position.

Servo Motor Pinout



Servo Motor Pinout



Arduino IDE Library

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Controlling Servo Motor

Required Elements:



Arduino

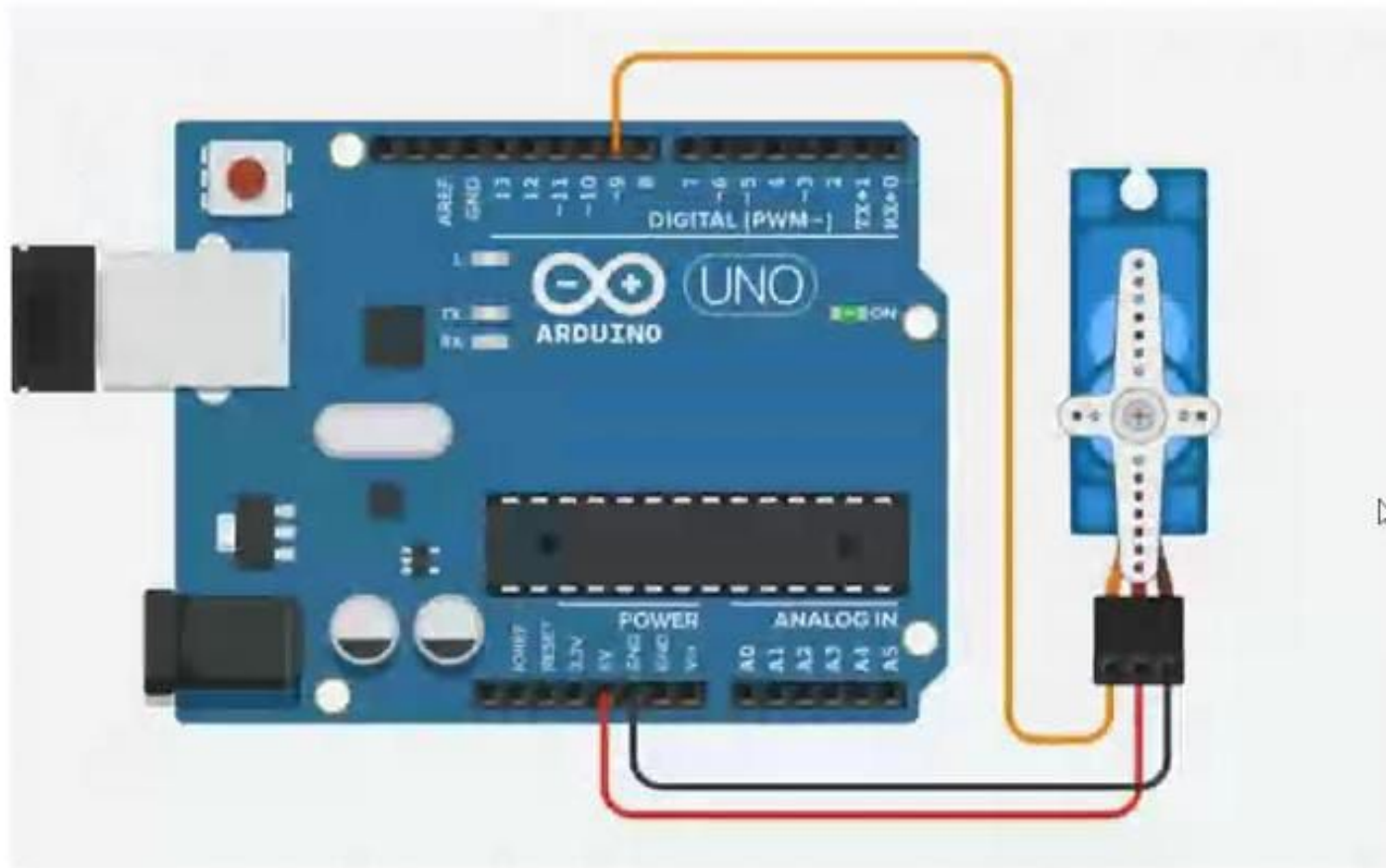


Servo Motor



Breadboard

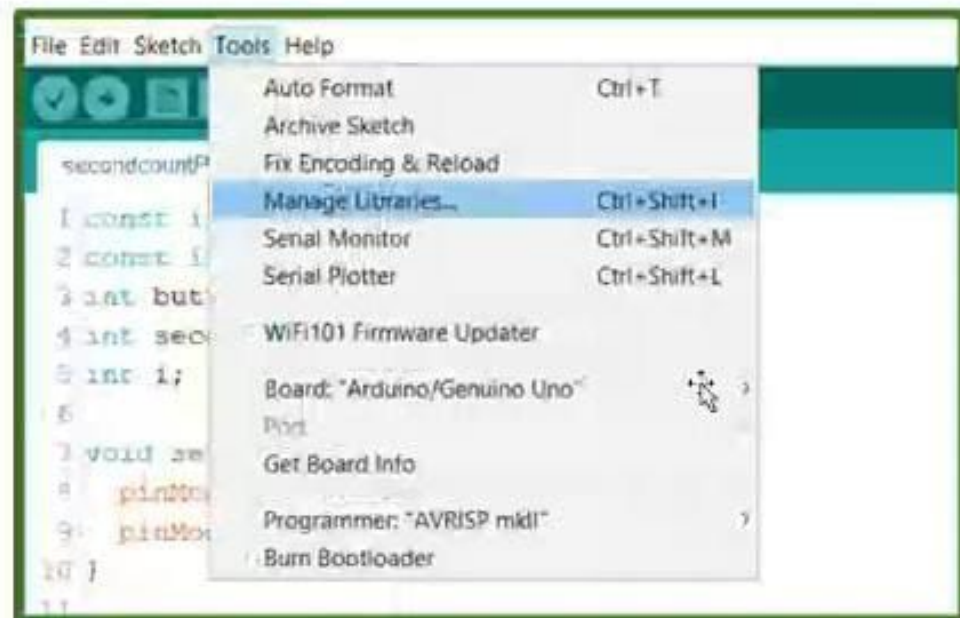
Controlling Servo Motor



28

Where to find these libraries?

In the Arduino IDE, we have a "manage Libraries" tool. This tool can be used to install any needed library for the common sensors. The benefit of this tool is that we no longer need to copy any file to the Arduino library, it is all done automatically.



Arduino IDE Library

Arduino V1.0.1

What is a library?

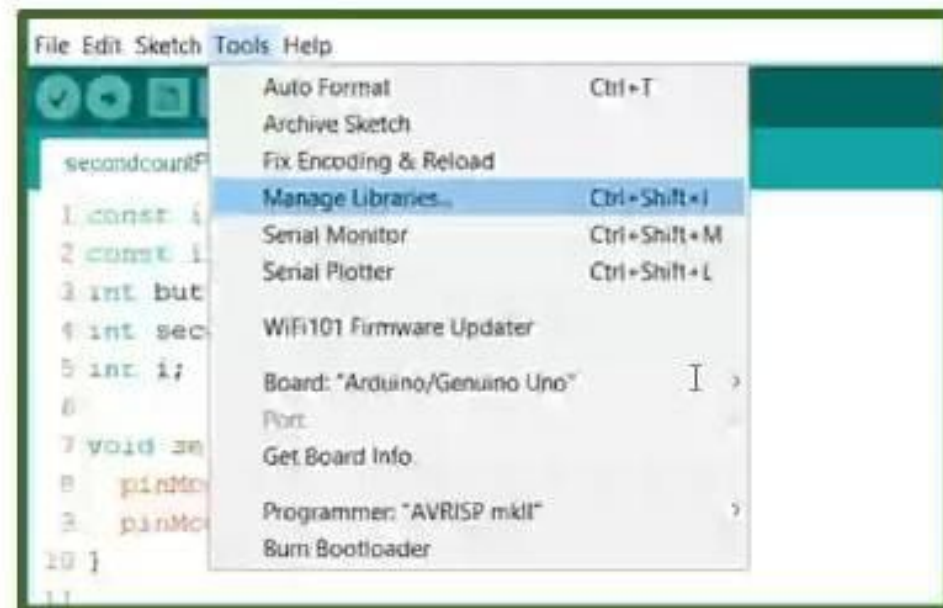
A library is a storage of functions. It has files that contain codes which can be used to run your sensors. Since each sensor needs special input or output, we need to use a special program to transform the information from the sensor to the user.

Let's consider an Ultrasonic sensor, which requires two digital pins, a VCC and GND pin. Each of these pins should be defined. Then a series of codes should be applied. This process can be reduced if we use a Library for that ultrasonic sensor. This library will simplify the main codes of the ultrasonic sensor and will represent them in functions.



Where to find these libraries?

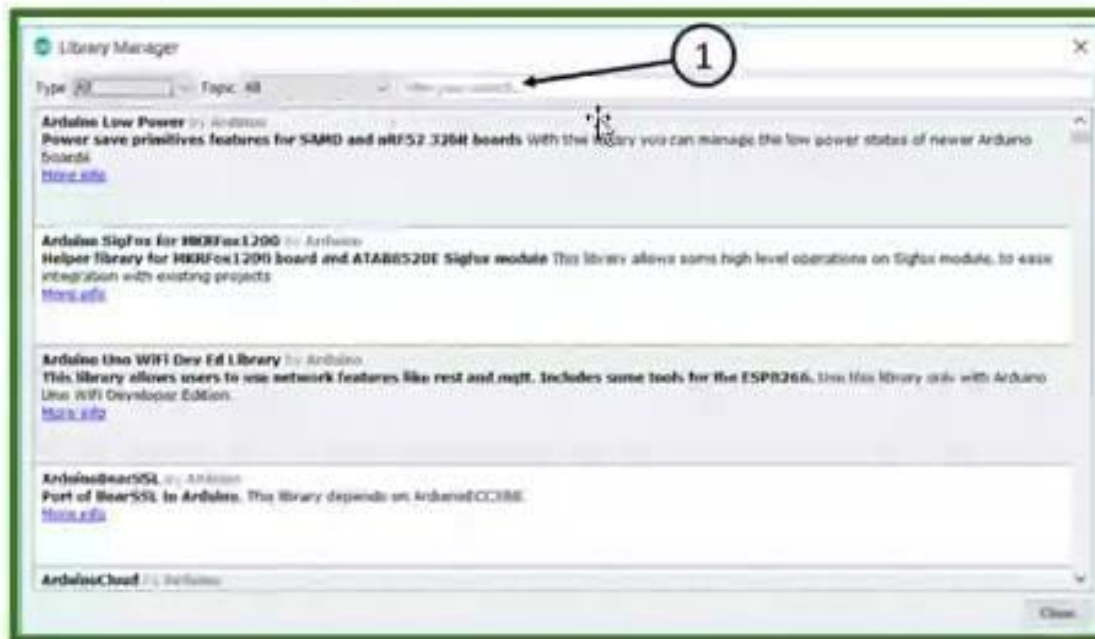
In the Arduino IDE, we have a "manage Libraries" tool. This tool can be used to install any needed library for the common sensors. The benefit of this tool is that we no longer need to copy any file to the Arduino library, it is all done automatically.



Where to find these libraries?

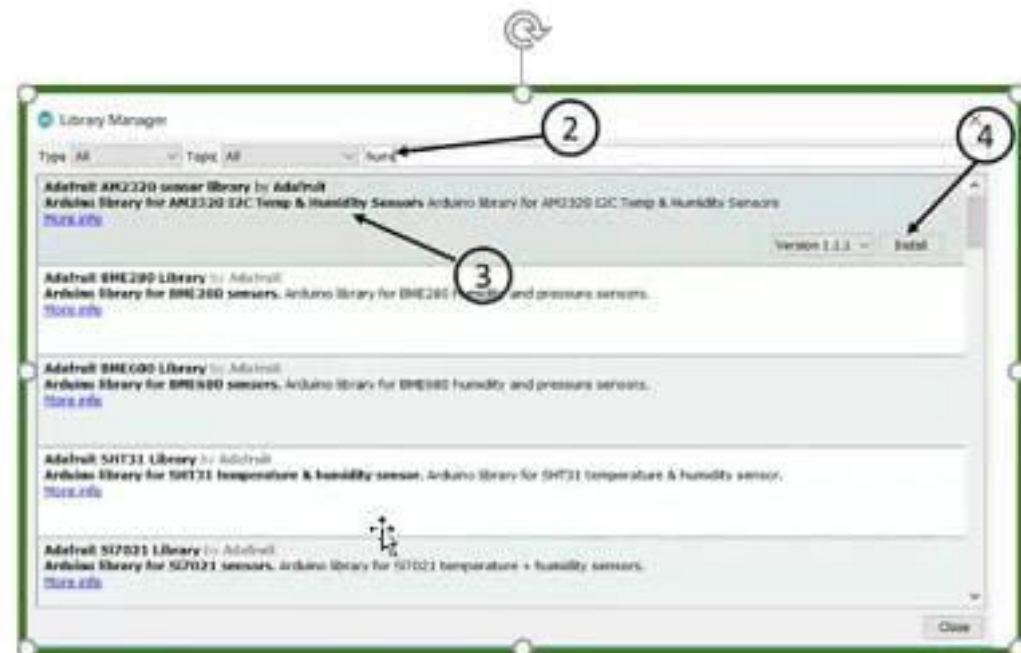
Once you open the Library manager go to

- 1- Search bar, where you can type your sensor name or number



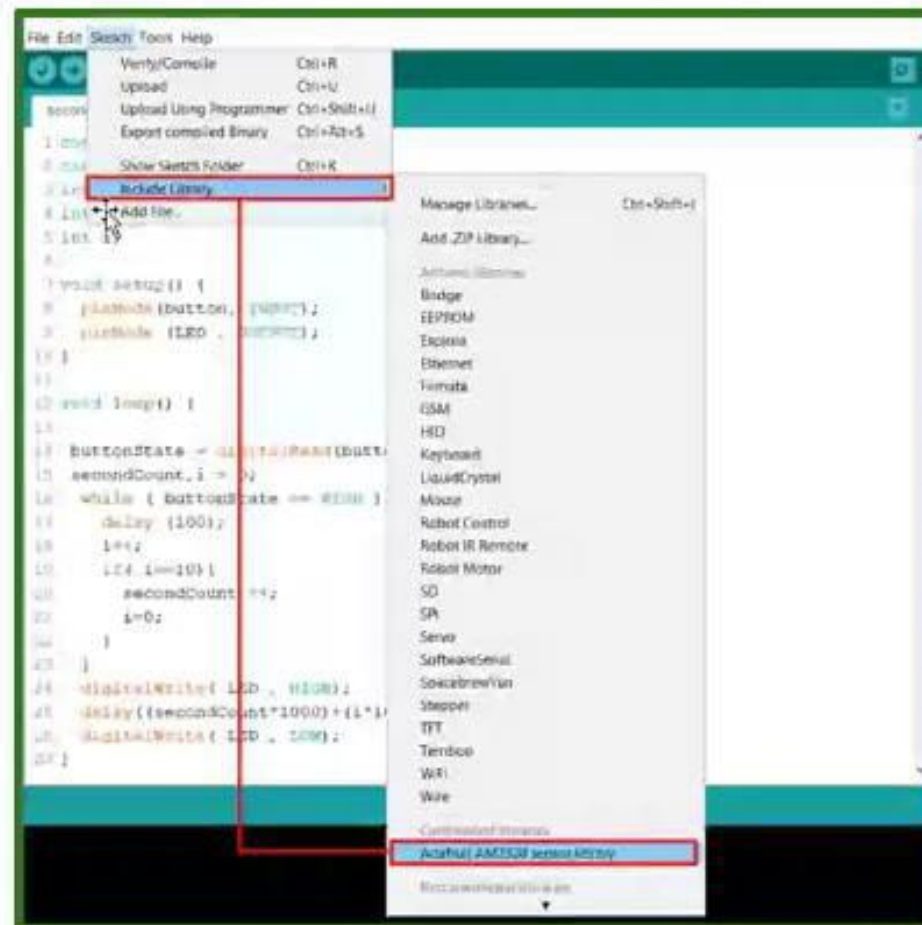
Where to find these libraries?

- 2- Write humidity sensor
- 3- Check the number of the sensor
- 4- Install the library



Accessing your Library

- 1- Navigate to sketch
- 2- Include Library
- 3- Include your library.



- Verify/Compile Ctrl+R
- Upload Ctrl+U
- Configure and Upload
- Upload Using Programmer Ctrl+Shift+U
- Export Compiled Binary Alt+Ctrl+S
- Optimize for Debugging

Show Sketch Folder Alt+Ctrl+K

Include Library >

Add File...

Manage Libraries... Ctrl+Shift+I

Add .ZIP Library...

Arduino libraries

Arduino_BuiltIn

EEPROM

Ethernet

Firmata

HID

Keyboard

LiquidCrystal

Mouse

SD

SoftwareSerial

SPI

Stepper

TFT

Wire

Contributed libraries

Adafruit Motor Shield library

Adafruit Unified Sensor

AM2320_asukiaaa

DHT sensor library

DHT12 sensor library

IRremote

LCD_I2C

NewPing

Servo

un once:

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Go to Settings to activate Windows.

Ln 10, Col 1

Controlling Servo Motor

